DDANN-2D (schwarz)

Problem 1 — 2D Sine-Sine (Zero BCs)

$$-\Delta u = 2\pi^2 \sin(\pi x)\sin(\pi y), \quad u = 0 \text{ on all 4 edges}$$

$$\text{Exact: } u(x,y) = \sin(\pi x)\sin(\pi y)$$

The solution is zero on all boundaries and peaks at $(0.5, 0.5)$ with value 1. Interface value at $x = 0.5$: $u(0.5, y) = \sin(\pi/2)\sin(\pi y) = \sin(\pi y)$ — a full sine wave in y.

```
Evaluation:
  Relative L2 Error  : 0.035016
  Max Absolute Error : 0.036228
  Mean Absolute Error: 0.014799
```

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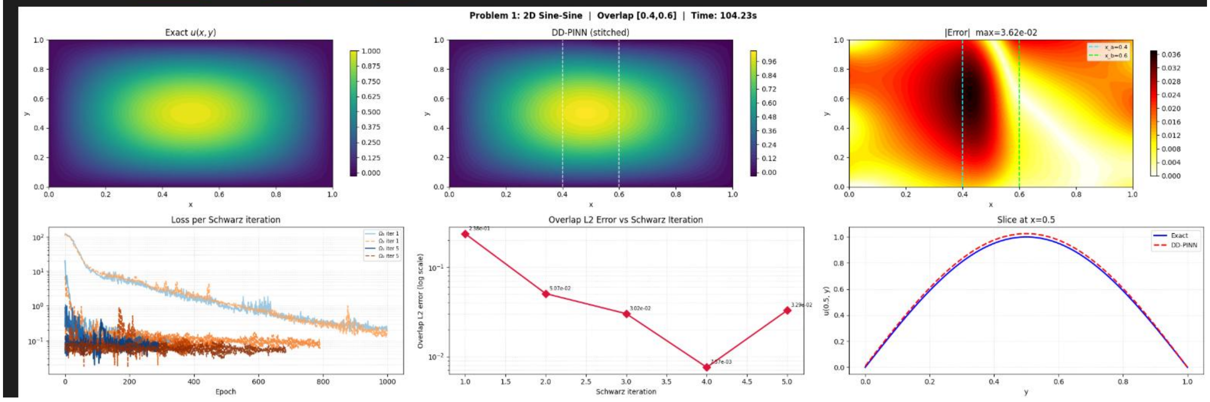

Problem 2 — 2D Polynomial (Zero BCs)

$$-\Delta u = 2[x(1-x) + y(1-y)], \quad u = 0 \text{ on all 4 edges}$$

$$\text{Exact: } u(x,y) = x(1-x)\cdot y(1-y)$$

Interface value at $x = 0.5$: $u(0.5, y) = 0.5 \cdot 0.5 \cdot y(1-y) = 0.25\, y(1-y)$ — a parabola in y.

```
...
Evaluation:
  Relative L2 Error  : 0.171542
  Max Absolute Error : 0.011999
  Mean Absolute Error: 0.004984
```

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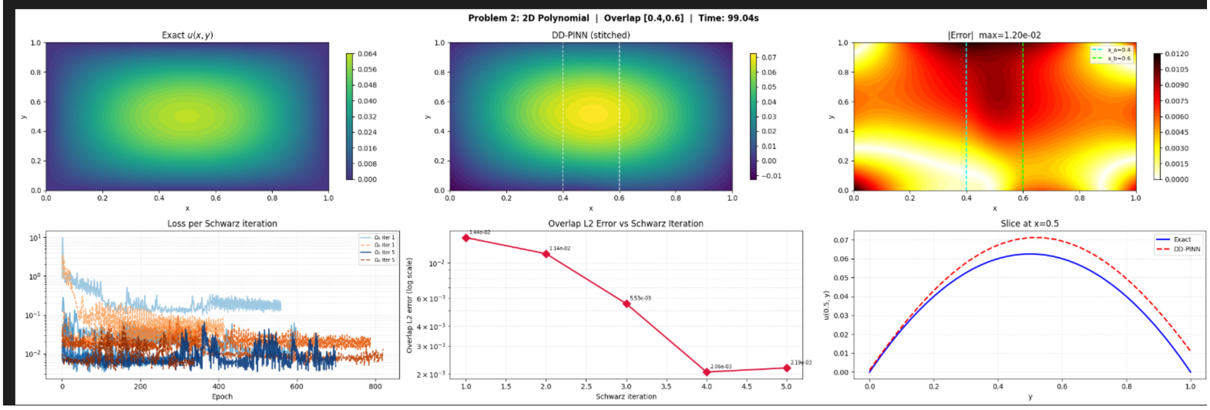

Problem 3 — 2D High-Frequency (Spectral Bias Demo)

$$-\Delta u = 8\pi^2 \sin(2\pi x)\sin(2\pi y), \quad u = 0 \text{ on all 4 edges}$$

$$\text{Exact: } u(x, y) = \sin(2\pi x)\sin(2\pi y)$$

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Evaluation:
  Relative L2 Error  : 0.020044
  Max Absolute Error : 0.037352
  Mean Absolute Error: 0.007623
```

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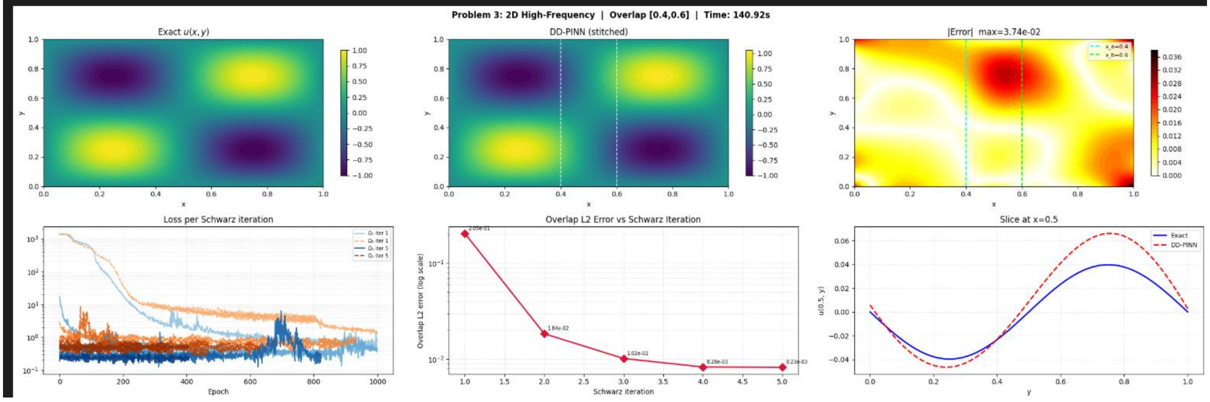

Problem 4 — 2D Exponential (Non-Zero BCs)

$$-\Delta u = -3e^{x+y}, \quad u = e^{x+y} \text{ on boundary}$$

$$\text{Exact: } u(x, y) = e^{x+y}$$

Non-zero BCs test whether the Schwarz method correctly handles non-homogeneous problems. Interface value at $x = 0.5$: $u(0.5, y) = e^{0.5+y}$ — an exponential in y.

Verification: $\partial^2 u/\partial x^2 = e^{x+y}$, $\partial^2 u/\partial y^2 = e^{x+y}$, so $-\Delta u = -2e^{x+y}$... Wait, actually: $-\Delta(e^{x+y}) = -(e^{x+y} + e^{x+y}) = -2e^{x+y}$... Let me use $-\Delta u = -3e^{x+y}$... actually for $u = e^{x+y}$: $u_{xx} = e^{x+y}$, $u_{yy} = e^{x+y}$, so $-\Delta u = -2e^{x+y}$. Using $u = e^{x+y}$ with $-\Delta u = -2e^{x+y}$.

```
Evaluation:
  Relative L2 Error  : 0.021440
  Max Absolute Error : 0.118272
  Mean Absolute Error: 0.060138
```

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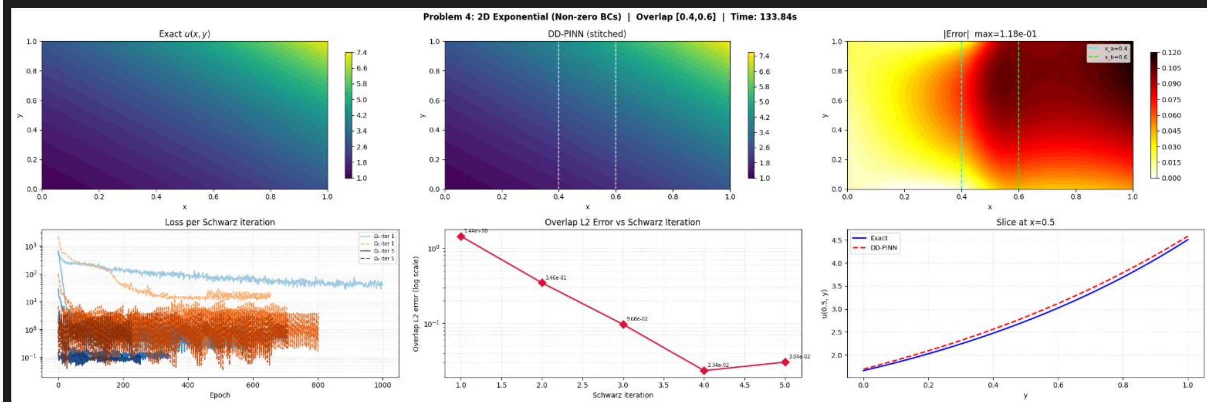

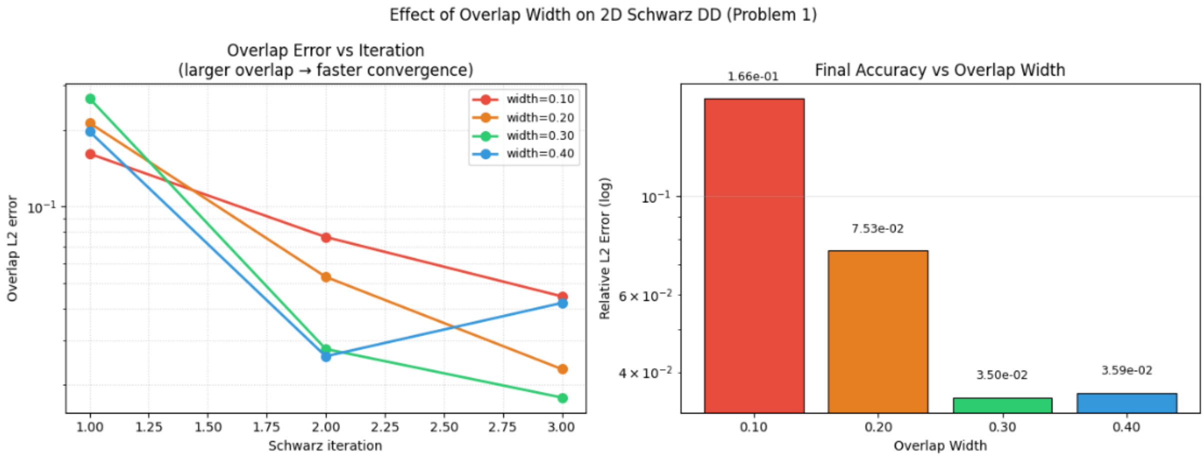

Effect of Overlap Width on 2D Schwarz DD (Problem 1)
Overlap Error vs Iteration
(larger overlap → faster convergence)
width=0.10
width=0.20
width=0.30
width=0.40
Overlap L2 error
10⁻¹
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Schwarz iteration
Final Accuracy vs Overlap Width
1.66e-01
7.53e-02
3.50e-02
3.59e-02
Relative L2 Error (log)
10⁻¹
6 × 10⁻²
4 × 10⁻²
0.10
0.20
0.30
0.40
Overlap Width